The ideal step in the energy representation

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Abstract

The Hamiltonian is written as the sum of the particle energies, $H = \sum_a e_a$, each evaluated from the particle's own entropy, momentum and volume. The volume is coarse grained with the entropy weights already in the code, $V_a = s_a/S(\boldsymbol{x}_a)$. Hamilton's equations then give an ideal step in which each particle moves with its own velocity and feels only a pressure force. Viscosity is left out.

I. VARIABLES AND THE VOLUME

Each particle a has a position $\boldsymbol{x}_a$, a momentum $\boldsymbol{m}_a$ and an entropy weight s_a. The weights are constants of the ideal step; a convenient start is $s_a = 1$ for all particles. The coarse-grained entropy density at the particle is the kernel sum

$$S_a \equiv S(\boldsymbol{x}_a) = \sum_b s_b\, W(\boldsymbol{x}_a - \boldsymbol{x}_b), \tag{1}$$

and the particle's volume is defined by

$$V_a = \frac{s_a}{S_a}, \qquad \text{so that} \qquad \frac{1}{V_a} = \frac{S_a}{s_a} \xrightarrow{s_a=1} \sum_b W(\boldsymbol{x}_a - \boldsymbol{x}_b). \tag{2}$$

For unit weights $1/V_a$ is the number density of particles. The particle's entropy density s_a/V_a is then the smoothed one, S_a. Its momentum density is its own, $\boldsymbol{m}_a/V_a$, not smoothed.

When the particles move, the volume changes as

$$\frac{\dot V_a}{V_a} = -\frac{\dot S_a}{S_a} = \frac{1}{S_a}\sum_b s_b\,(\dot{\boldsymbol{x}}_b - \dot{\boldsymbol{x}}_a)\cdot \boldsymbol{D}_{ab}, \qquad \boldsymbol{D}_{ab} \equiv \frac{\partial W(\boldsymbol{x}_a - \boldsymbol{x}_b)}{\partial \boldsymbol{x}_a}, \tag{3}$$

the SPH estimate of $\nabla\cdot\dot{\boldsymbol{x}}$ at $\boldsymbol{x}_a$.

II. THE PARTICLE ENERGY

The equation of state gives the energy density $h(S, \boldsymbol{M})$ as a function of the entropy and momentum densities. The particle's energy is

$$e_a = E_a(s_a, \boldsymbol{m}_a, V_a) \equiv V_a\, h\Big(\frac{s_a}{V_a}, \frac{\boldsymbol{m}_a}{V_a}\Big), \tag{4}$$

the same particle energy as in the particle representation of `changerep.tex`. Its differential is the first law for the particle,

$$de_a = T_a\, ds_a + \boldsymbol{v}_a\cdot d\boldsymbol{m}_a - p_a\, dV_a, \tag{5}$$

where T_a (bare, $T_{\rm rel}/\gamma$), $\boldsymbol{v}_a$ and p_a are the temperature, velocity and pressure at the particle's densities $(S_a, \boldsymbol{m}_a/V_a)$.

III. THE HAMILTONIAN AND HAMILTON'S EQUATIONS

The Hamiltonian is the sum of the particle energies,

$$\boxed{H(\boldsymbol{x},\boldsymbol{m}) = \sum_a e_a = \sum_a V_a\, h\Big(\frac{s_a}{V_a}, \frac{\boldsymbol{m}_a}{V_a}\Big), \qquad V_a = \frac{s_a}{S(\boldsymbol{x}_a)}} \tag{6}$$

It depends on the positions only through the volumes. Since $s_a = V_a S_a$, eq. (6) is the same as

$$H = \sum_a s_a\, \mathcal{E}\Big(S_a, \frac{\boldsymbol{m}_a}{s_a}\Big), \tag{7}$$

with $\mathcal{E}$ the energy per unit entropy. The present Hamiltonian (`sph2.tex`) is $\sum_a s_a \mathcal{E}(S_a, \boldsymbol{M}_a/S_a)$, with the smoothed momentum $\boldsymbol{M}_a = \sum_b \boldsymbol{m}_b W_{ab}$ in place of the particle's own $\boldsymbol{m}_a$; that is the only difference.

Hamilton's equations are

$$\dot{\boldsymbol{x}}_a = \frac{\partial H}{\partial \boldsymbol{m}_a}, \qquad \dot{\boldsymbol{m}}_a = -\frac{\partial H}{\partial \boldsymbol{x}_a}, \qquad \dot{s}_a = 0. \tag{8}$$

By eq. (5), $\partial H/\partial \boldsymbol{m}_a = \boldsymbol{v}_a$: the particle moves with its own velocity,

$$\boxed{\dot{\boldsymbol{x}}_a = \boldsymbol{v}_a} \tag{9}$$

For the force, $\partial H/\partial \boldsymbol{x}_a = -\sum_c p_c\, \partial V_c/\partial \boldsymbol{x}_a$ with $\partial V_c/\partial \boldsymbol{x}_a = -(V_c/S_c)\, \partial S_c/\partial \boldsymbol{x}_a$, and

$$\frac{\partial S_c}{\partial \boldsymbol{x}_a} = \delta_{ca} \sum_b s_b \boldsymbol{D}_{ab} + s_a \boldsymbol{D}_{ac}. \tag{10}$$

Collecting terms,

$$\boxed{\dot{\boldsymbol{m}}_a = -\sum_b s_a s_b \left(\frac{p_a}{S_a^2} + \frac{p_b}{S_b^2}\right) \boldsymbol{D}_{ab}} \tag{11}$$

This is the standard symmetric SPH pressure force, with S in the role of the mass density and s_a in the role of the mass. Compared with `sph2.tex`, the terms in $\Delta\mathcal{M}_b(\boldsymbol{x}_a) = \boldsymbol{m}_b/s_b - \boldsymbol{M}_a/S_a$ are gone, because the momentum is not smoothed.

a. Continuum limit. With $\sum_b V_b f_b \boldsymbol{D}_{ab} \to \nabla f$, eq. (11) becomes $\dot{\boldsymbol{m}}_a \to -V_a \nabla p$. This is the momentum equation for a parcel moving with $\boldsymbol{v}$: relative to such a parcel the momentum flux $T^{ij} = M^i v^j + p\,\delta^{ij}$ reduces to $p\,\delta^{ij}$.

b. The particle energies. From eq. (5) at fixed s_a,

$$\dot{e}_a = \boldsymbol{v}_a \cdot \dot{\boldsymbol{m}}_a - p_a \dot{V}_a, \tag{12}$$

with $\dot{V}_a$ from eq. (3). The particle energies change by the work done on the particle; their sum does not.

IV. WHAT IS CONSERVED

1. *Energy.* $H = \sum_a e_a$ is conserved by Hamilton's equations, and it is the same sum of particle energies eq. (4) that the viscous step of `changerep.tex` works with: the smoothed and the particle energies are one and the same.

2. *Momentum.* $\boldsymbol{D}_{ba} = -\boldsymbol{D}_{ab}$, so the pair forces in eq. (11) cancel and $\sum_a \boldsymbol{m}_a$ is conserved.

3. *Entropy.* Each s_a is constant, so $\sum_a s_a$ is conserved, and each particle's entropy $S_a(e_a, \boldsymbol{m}_a)$ at volume V_a equals s_a.

V. WHAT IT DOES NOT CHANGE: THE STATIC RIGIDITY

At rest, $\boldsymbol{m}_a = 0$, the new and the present Hamiltonians are the same function of the positions,

$$H\big|_{\boldsymbol{m}=0} = \sum_a s_a\, \mathcal{E}(S_a, 0), \tag{13}$$

because the two differ only in which momentum enters $\mathcal{E}$. The static phonon spectrum of `rigidity.tex` (Sect. "The ideal Hamiltonian's phonons"), and with it the static shear modulus G and c_t, are therefore unchanged: the positions still enter only through the kernel sum $S_a = \sum_b s_b W_{ab}$, which is what breaks the relabeling symmetry. The energy representation is *not* a fix for the rigidity. (Volumes that partition space, $\sum_a V_a = A$, would be; that is a different choice of V_a in eq. (2), and eq. (6)–eq. (11) would carry over with $\partial V_c/\partial \boldsymbol{x}_a$ from the partition.)

What does change is the dynamics: the particles move with their own velocities rather than a doubly smoothed one, and the force has no $\Delta\mathcal{M}$ terms. Whether that changes the measured transverse oscillation in C_T (its frequency or damping) is an open question for the runs, not something this note predicts. The gain that is certain is consistency with the viscous step, Sect. VI.

VI. THE DETERMINISTIC VISCOUS STEP

With $H = \sum_a e_a$, the viscous step is done on the particles' own variables; there is no change of representation. The derivative of H with respect to a weight at fixed $\boldsymbol{x}$, $\boldsymbol{m}$ is

$$T_a^{\text{eff}} \equiv \frac{\partial H}{\partial s_a} = T_a - \frac{p_a}{S_a} + \sum_b s_b \frac{p_b}{S_b^2} W_{ab} = T_a + O(h^2), \tag{14}$$

using $\mathcal{E} - \boldsymbol{v}\cdot\boldsymbol{m}/s = T - p/S$; the last two terms are the smoothed minus the local p/S. With

$$\boldsymbol{\beta}_a = \frac{\boldsymbol{v}_a}{T_a^{\text{eff}}}, \qquad \Delta\boldsymbol{m}_a = -\Delta t\, \frac{\partial K}{\partial \boldsymbol{\beta}_a}, \qquad \Delta s_a = -\boldsymbol{\beta}_a \cdot \Delta\boldsymbol{m}_a, \tag{15}$$

$\Delta H = \sum_a (T_a^{\text{eff}} \Delta s_a + \boldsymbol{v}_a \cdot \Delta\boldsymbol{m}_a) = 0$ to first order in Δt, and $\sum_a \Delta s_a = 2K\Delta t \geq 0$, as in `sph2.tex`. K is the bond dissipative function of `changerep.tex`, which already takes a pointwise $\boldsymbol{\beta}_a$.

Compared with `changerep.tex`: the forward map $s_a \to e_a$ is no longer a map, since e_a is already the energy in H; the gradient correction g_a (its Appendix A) was the difference

between $\sum_a e_a$ and the smoothed H, and is now zero. The exact back map at fixed e_a, V_a does not carry over as it is: a change of s_a changes S_b, hence V_b, at every neighbour, so fixing every e_a at fixed V_a does not fix H exactly. The step eq. (15) conserves H to first order in Δt; the drift is measured (plan, step 6).

VII. IMPLEMENTATION PLAN

The Hamiltonian is chosen by a new field `params.hamiltonian` $\in$ {`:smoothed`, `:energy`} (environment `SPH_HAMILTONIAN`), default `:smoothed`, so that every existing run and test is unchanged.

a. Unchanged. The `Particle` struct (`x`, `s`, `m`, `S_smooth` and the EOS fields are enough); kernels, cell and neighbour lists; the EOS solvers `eos_newton`, `solve_eos!`, `particle_energy`, `eos_from_energy`; the integrators `leapfrog_step!` and `kdk_step!` (their structure; they call the functions below); the bond dissipative function and force; the set-up of initial conditions ($s_a = S(\boldsymbol{x}_a)\,\mathrm{vol}$, or $s_a = 1$); the recorders. The stochastic step is left alone.

b. Changed (all in `celllist.jl`, dispatched on `params.hamiltonian`).

1. `smooth_to_particles!`: sum S_a only, and set the momentum density to the particle's own, `M_smooth` $= \boldsymbol{m}_a S_a/s_a = \boldsymbol{m}_a/V_a$. Then `solve_eos!` returns, unchanged, the particle's own T_a, $\boldsymbol{v}_a$, p_a and e_a/s_a (`H_over_S`). One neighbour sum fewer.

2. `update_velocities!`: $\dot{\boldsymbol{x}}_a = \boldsymbol{v}_a$, local, no neighbour sum; `Teff` from eq. (14), one neighbour sum (needed only by the viscous step and by the bond metric).

3. `hamiltonian_force!`: the pair bracket becomes $-s_a s_b(p_a/S_a^2 + p_b/S_b^2)$, eq. (11).

4. Viscous step: `dissipative_step_changerep!` with $\boldsymbol{\beta}_a$ of eq. (15) read from the particle (no `pointwise_state` copy), and $\Delta s_a = -\boldsymbol{\beta}_a \cdot \Delta \boldsymbol{m}_a$. `dissipative_step!` calls it, not the gradient-corrected step, when `params.hamiltonian == :energy`.

5. Diagnostics: the total energy is $\sum_a e_a = \sum_a s_a\,\mathtt{H_over_S}_a$, the same number as H.

c. Steps and checks.

1. Implement 1–3 behind the flag; `test/testregress_step.jl` must stay bit-identical for `:smoothed`.

2. A test `test/testenergystep.jl`: $\partial H/\partial \boldsymbol{m}_a$ and $\partial H/\partial \boldsymbol{x}_a$ against finite differences of H on a jittered lattice; $\sum_a \dot{\boldsymbol{m}}_a = 0$; $\dot{s}_a = 0$.

3. Ideal runs: energy drift of H versus Δt (second order for the leapfrog), a sound wave (speed and damping) against the present step (`examples/testsound.jl`), and the time step: the unsmoothed $\boldsymbol{v}_a$ may need a smaller CFL.

4. Static check: the phonon spectrum of Sect. V must agree with the present one to roundoff (it is the same H at $\boldsymbol{m} = 0$).

5. Implement 4; check $\sum_a \Delta s_a = 2K\Delta t$ and $\Delta H = O(\Delta t^2)$ per step; a shear wave against $e^{-\nu k^2 t}$ (`examples/testshearwave.jl`).

6. Energy drift over long coupled deterministic runs, against the present step with the gradient-corrected map.

d. Documentation. Each changed function gets a comment that states the equation it implements and cites this note by equation; `params.hamiltonian` is described where `SimParams` is defined; the outcome of each check above is added to this note as it is done.

e. Status (2026-11-09). Items 1–3 of the changes and item 4 (the viscous step, `dissipative_step_energy!`) are in `src/sph.jl` and `src/celllist.jl`. On a jittered lattice with a moving fluid, $\partial H/\partial \boldsymbol{m}_a$, $\partial H/\partial \boldsymbol{x}_a$ and $\partial H/\partial s_a$ agree with $\dot{\boldsymbol{x}}_a$, $-\dot{\boldsymbol{m}}_a$ and T_a^{eff} to 10^{-6} (finite differences); $\sum_a \dot{\boldsymbol{m}}_a = 7 \times 10^{-16}$; $H = \sum_a$`particle_energy` exactly. The deterministic shear wave is Sect. VIII.

VIII. THE DETERMINISTIC SHEAR WAVE

`examples/shearwave_decay.jl` (viscous and ideal) and `examples/shearwave_scan.jl` (ideal, kernels and h/a). A uniform fluid with

$$M_y(x,0) = A \sin kx, \qquad k = \frac{2\pi}{L}, \tag{16}$$

evolved with the full deterministic step. A fluid keeps the amplitude with $\eta = 0$ and decays as $e^{-\nu k^2 t}$, $\nu = (\eta/s)/T_{\mathrm{rel}}$, with $\eta > 0$. An elastic solid with transverse speed c_t instead gives, with $\eta = 0$,

$$\frac{A(t)}{A(0)} = \cos(c_t k t). \tag{17}$$

At $h/a = 2$ (C2, $kh = 0.31$):

1. The two Hamiltonians agree: amplitudes to 10^{-3} with and without viscosity, $|\Delta H/H| \sim 10^{-11}$ for the ideal step. The viscous step drifts by 4×10^{-8} (`:energy`) and 8×10^{-9} (`:smoothed`).

2. For $A \le 0.005$ (displacements $\le 0.4a$) the ideal wave is an undamped cosine with $c_t = 0.0340$ (`:energy`) and 0.0335 (`:smoothed`): the static value 0.034 of `rigidity.tex`. For $A = 0.02$ the particles slip by $2.4a$ and the response is not linear.

3. With $\eta/s = 0.02$ the amplitude falls below $e^{-\nu k^2 t}$ and turns negative: a damped transverse oscillation.

The scan over kernels and h/a is running.

IX. PROPOSAL: VOLUMES THAT ADD UP TO THE AREA

a. Why the kernel volume gives a modulus. Since H depends on the positions only through the volumes, to second order in their changes

$$\delta H = -\sum_a p_a\,\delta V_a + \sum_a \frac{K_a}{2V_a}\,\delta V_a^2, \tag{18}$$

with K the bulk modulus. For $V_a = s_a/S_a$ a homogeneous shear ϵ of a lattice changes every volume by the same amount at second order, $\delta V_a = c\,\epsilon^2$, so $\sum_a \delta V_a \neq 0$, and the first term gives a modulus

$$G \sim p\,c(h/a, \text{kernel}). \tag{19}$$

This is the ϵ of `rigidity.tex`, proportional to the pressure.

b. The condition. If the volumes always add up to the area,

$$\sum_a V_a = A, \tag{20}$$

then at uniform pressure $\sum_a p\,\delta V_a = p\sum_a \delta V_a = 0$, and the second term of eq. (18) is of fourth order in ϵ on a lattice: no modulus. A partition of space (Voronoi cells, or the partition of unity of meshless finite-mass methods) satisfies eq. (20); so does the simplest choice, a global normalization of the kernel volume,

$$V_a = \frac{A}{Z}\,\tilde V_a, \qquad \tilde V_a = \frac{s_a}{S_a}, \qquad Z = \sum_c \tilde V_c. \tag{21}$$

The particle's entropy density is then $s_a/V_a = (Z/A)\,S_a$.

c. Hamilton's equations. With eq. (21) in eq. (6), $\partial H/\partial \boldsymbol{x}_a = -\sum_c p_c\,\partial V_c/\partial \boldsymbol{x}_a$ and

$$\frac{\partial V_c}{\partial \boldsymbol{x}_a} = \frac{A}{Z}\Big(\frac{\partial \tilde V_c}{\partial \boldsymbol{x}_a} - \frac{\tilde V_c}{Z}\sum_d \frac{\partial \tilde V_d}{\partial \boldsymbol{x}_a}\Big). \tag{22}$$

The second term subtracts the volume-averaged pressure

$$\bar p = \frac{1}{A}\sum_c p_c V_c, \tag{23}$$

and the force is eq. (11) with $p \to p - \bar p$:

$$\dot{\boldsymbol{m}}_a = -\frac{A}{Z}\sum_b s_a s_b \left(\frac{p_a - \bar p}{S_a^2} + \frac{p_b - \bar p}{S_b^2}\right)\boldsymbol{D}_{ab}. \tag{24}$$

The same substitution in eq. (14) gives the derivative with respect to the weights,

$$T_a^{\text{eff}} = T_a - \frac{A}{Z}\left[\frac{p_a - \bar p}{S_a} - \sum_b s_b\,\frac{p_b - \bar p}{S_b^2}\,W_{ab}\right]. \tag{25}$$

Still exact: H is conserved by Hamilton's equations, and the pair terms of eq. (24) are odd under $a \leftrightarrow b$, so momentum is conserved. The continuum limit is unchanged, $\nabla(p - \bar p) = \nabla p$. The cost is one global sum per step. In SPH, subtracting a background pressure is a known remedy for the tensile instability; here it follows from the Hamiltonian, and energy conservation is kept.

d. Expected, and tests.

1. Perfect lattice: c_t falls from 0.034 to near zero. Test: the shear-wave scan with eq. (21), against Sect. VIII.

2. Thermal state: what is left is driven by the pressure fluctuations δp rather than by p, with random sign; no coherent long-wavelength modulus is expected. Test: the stochastic runs, Sect. X.

3. Disordered arrangements: an affine shear changes the individual δV_a at first order, with random sign, which leaves a modulus $\propto K\,(\text{disorder})^2$ from the second term of eq. (18): the caveat (a) of `rigidity.tex` for Voronoi volumes. A local partition does not remove it either. Test: the static phonon spectrum on jittered lattices (`ideal_lattice_modes.jl`, `JITTER`).

Implementation: an option `volume = :normalized` of the `:energy` Hamiltonian; one reduction for $\bar{p}$ and Z, and $p \to p - \bar{p}$, $S \to (Z/A)S$ where the particle state is evaluated.

X. THE STOCHASTIC STEP WITH `:ENERGY`

Not yet tested. The stochastic step (`stochrep.tex`) works at fixed positions, volumes $V_a = s_a/S_a$ and particle energies e_a, which are exactly the variables of eq. (6); on a frozen lattice it is the same step for both Hamiltonians. Two differences:

1. *No gradient correction.* The gradient energy g_a of `changerep.tex` (Appendix A) absorbs the difference between $\sum_a e_a$ and the smoothed H, which is now zero. With `:energy` the step must use the plain back map $s'_a = S_a(e_a, \boldsymbol{m}'_a)$ (`gradient = false`).

2. *The back map and the volumes.* The back map holds e_a and V_a fixed, but a change of s_a changes S_b and so V_b at every neighbour. To first order

$$\Delta H \simeq \sum_a \left(T_a^{\text{eff}} - T_a\right)\Delta s_a, \tag{26}$$

with $T_a^{\text{eff}} - T_a$ the smoothed minus the local p/S, of order h^2. In the deterministic step this is 4×10^{-8} (Sect. VIII); with noise the Δs_a are larger and fluctuate, so its size has to be measured.

Tests, in order:

1. `stochrep_step!` defaults to `gradient = false` when `params.hamiltonian == :energy`, and the drivers keep that default.

2. One step on a thermal state: ΔH per stochastic step, `:energy` against `:smoothed` (with its gradient-corrected map).

3. Coupled runs, `stochrep_hamiltonian.jl`: $L = 0.6$, $h/a = 2$, $\eta/s = 0.2$, $S_0 = 10^6$, to $t \approx 50$, 2–4 realizations per Hamiltonian: $\Delta H/H(t)$, KE/eq, the lattice disorder. Local.

4. At NERSC, after 1–3: equilibrium C_T, C_L with `:energy`, as campaign `second`; the node-hour estimate first.

a. Status. Test 1 is done; `:smoothed` stays bit-identical (`test/testregress_step.jl`). Test 2 (`examples/energystep_stoch_onestep.jl`, $L = 0.6$, $h/a = 2$, equilibrium start, 20 half steps $\Delta t/2 = 0.0043$ at fixed positions, 3 seeds):

	`:smoothed` (gradient map)	`:energy`
$\Delta H/H$ per half step, rms	3.4×10^{-5}	1.8×10^{-7}
$\Delta H/H$ per half step, mean	$\pm 7 \times 10^{-6}$ (seed to seed)	-1.8×10^{-7} (every seed)

With `:energy`, $\Delta H = \Delta \sum_a e_a$ exactly, and the change is 200 times smaller than with `:smoothed`, but systematic: a cooling of about 4×10^{-5} per unit time. It is correlated (-0.8) with $\sum_a (T_a^{\text{eff}} - T_a)\Delta s_a$, but of the opposite sign and twice the size; not understood yet.

With Sect. IX, the same tests with `volume = :normalized` decide whether the transverse oscillation of the equilibrium C_T is the pressure term of eq. (18).